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Alessandro Pedone
Curriculum Vitae

PERSONAL DATA

Date of Birth	01/10/2002
Nationality	Italian
Place of Birth	Milan, Italy

CURRENT POSITION

Politecnico di Milano — Pursuing Master's Degree in Mathematical Engineering, track in Computational Science and Computational Learning

EDUCATION

2024 - Current — **Master's Degree in Mathematical Engineering, Politecnico di Milano**

2021 - 2024 — **Bachelor's Degree in Mathematical Engineering, Politecnico di Milano**

Final grade: 110/110 cum laude

Thesis: The Cauchy-Kowalevski theorem and some of its consequences

Keywords: PDEs, characteristics method, analyticity/holomorphy, power series, method of majorants, Cauchy-Kowalevski, Holmgren and Cartan-Kähler theorems

Supervisor: Prof. Maurizio Grasselli

EXPERIENCE

01/12/2025 - 03/12/2025 — **SISSA Junior Math Days**

Event organized every year by SISSA, with the goal of allowing Master's students in their final year to get in touch with the mathematical research environment (professors and PhD students) at SISSA.

22/09/2025 - 26/09/2025 — **Workshop and Summer School on Applied Analysis 2025**

Organized by TU Chemnitz.

Topics:
- High-Dimensional Approximations for Parametric and Random PDEs
- Inverse Optimal Transport Introduction
- Proximal Optimization and Deep Learning Methods (with applications to designing networks architectures)
- Data-Driven Methods in Control (focus on Koopman-based techniques)
- The Approximation Theory of (Shallow and Deep) Neural Networks

23/06/2025 - 27/06/2025 **Theorical Foundations of Machine Learning 2025**

Admitted (as a Master's student) to a selective PhD program organized by MaLGa (Machine Learning Genoa Center) at the University of Genoa, with an acceptance rate of approximately 35%, primarily comprising PhD candidates.

Topics:
- Statistical learning theory framework
- Kernel methods and neural networks
- Empirical risk minimization and regularization
- Reproducing kernel Hilbert spaces (RKHS)
- Optimization techniques: convex analysis, gradient methods, stochastic optimization, splitting methods, backpropagation
- Theoretical analysis: concentration inequalities, empirical process theory, spectral calculus, operator theory, Rademacher complexities

PROJECTS

04/2025 - Current **Geom-DeepONet (in progress)**

Reference: 🔗

Objective: develop a surrogate model for the electrostatic problem in an idealized MEMS with varying geometry.

Future applications: coupling the surrogate model to solve multiphysics (electro-mechanical) problems.

04/2025 - Current **Coupling MPE and Stokes' equations for cerebrospinal fluid flow and tissue motion (in progress)**

Reference: 🔗

Objective: study cerebrospinal fluid (CSF) flow, inside brain ventricles and subarachnoid space, with a focus on modeling the latter and exploiting already existing results.

Methodology:
1. *Implementation challenges:* segmenting the geometry (obtained from MRI) and creating the mesh of the subarachnoid space.
2. *Mathematical model:* coupling of MPE for brain tissue poromechanics Stokes' equations for cerebrospinal fluid flow.
3. *Numerical tools:* high order discontinuous Galerkin method on polytopal meshes for spatial discretization.

03/2025 - 05/2025 **C++ Scientific Computing Projects**

1. Some optimization methods for real-valued functions (Gradient Descent, Heavy Ball, Nesterov, ADAM...)
2. Parallel implementations of two kind of solvers for the Poisson equation in the unit square (Schwarz method)
3. Implementation of a matrix class, particularly suited for sparse matrices (CSR/CSC formats and modified formats)

12/2024 **Mars Terrain segmentation**

Objective: segment Mars terrain images into five classes (Background, Soil, Bedrock, Sand, and Big Rock) using a UNet architecture.

Dataset: 2,615 grayscale images (64x128 resolution), filtered to 2,505 images.

11/2024 **Blood Cell Classification**

Objective: perform an 8-class classification task on blood cells images using CNNs.

Dataset: 13,759 RGB images (96x96 resolution), filtered to 11,959 images after preprocessing.

04/2024 – 06/2024 **Statistical Inference for Math Performance**

Objective: build a linear regression model for math perfomance of italian high school students.

Dataset: OECD PISA 2022 survey.

SKILLS

Programming Languages	Python, C/C++, MATLAB, R, LaTeX
Libraries	NumPy, SciPy, scikit-learn, TensorFlow, Keras, Pandas, Matplotlib, Seaborn, MIP, FEniCS, FEniCSx
Operating Systems	Linux (Ubuntu), Windows
Additional Software	Notion
Memory techniques	Method of loci, Mnemonic major system, Mnemonic link system

LANGUAGES

Italian	Native speaker
English	C1 (IELTS 7.5)

HONOURS AND AWARDS

25/01/2023 — **Premio Migliori Matricole aa 21-22 Classe L 8 Fondo Giovani**

Best Freshmen Award, academic year 2021–2022, Class L-8, Fondo Giovani — granted upon completion of the first year of enrollment, based on academic performance, including GPA and the number of ECTS credits earned.

MEMBERSHIPS

2025 - Current	SIAM (Society for Industrial and Applied Mathematics)
2021 – Current	AIM (Associazione Ingegneri Matematici)

VOLUNTEERING

2020 - Current	Blood donor
2018	ABCDigital: Liceo Scientifico Elio Vittorini promoted a project to organize a free course for senior citizens, consisting of 10 weekly lectures (2 hours each), aimed at improving their digital skills, device usage, and online awareness.